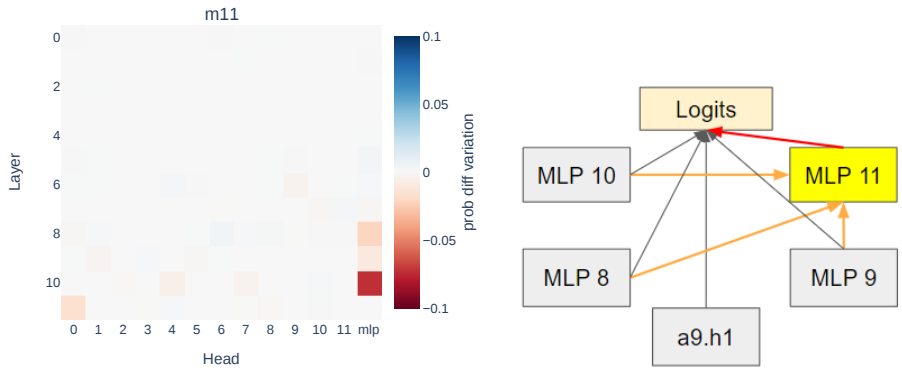

Figure 16: Path Patching Step 2: MLP 11

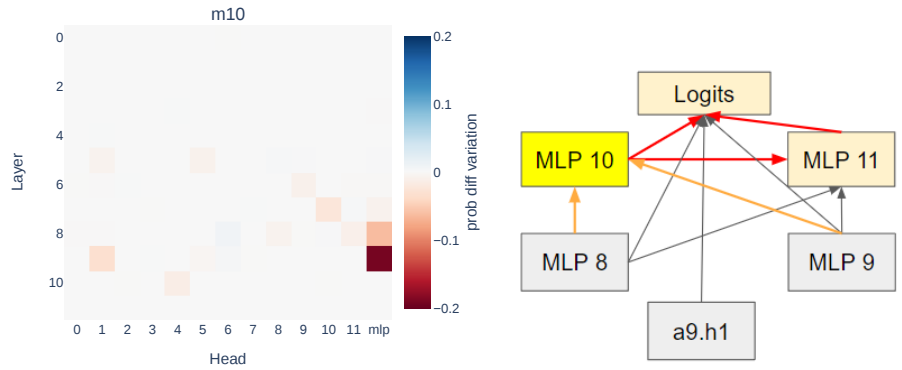

Figure 17: Path Patching Step 3: MLP 10

logits), in red (Figure 16, right). The results (Figure 16, left) indicate that the other MLPs influence the circuit most through MLP 11.

We continue onward to MLP 10, and see which nodes are most influential on the circuit via it. We consider all the ways in which nodes could contribute via MLP 10 to the circuit, shown in red on the right of Figure 17. The results (Figure 17) again that MLP 10 relies mostly on MLPs 8 and 9. We proceed similarly with MLP 9, considering all the ways in which it influences the circuit; the results in Figure 18 indicate that it relies mostly on MLP 8 and a9.h1.

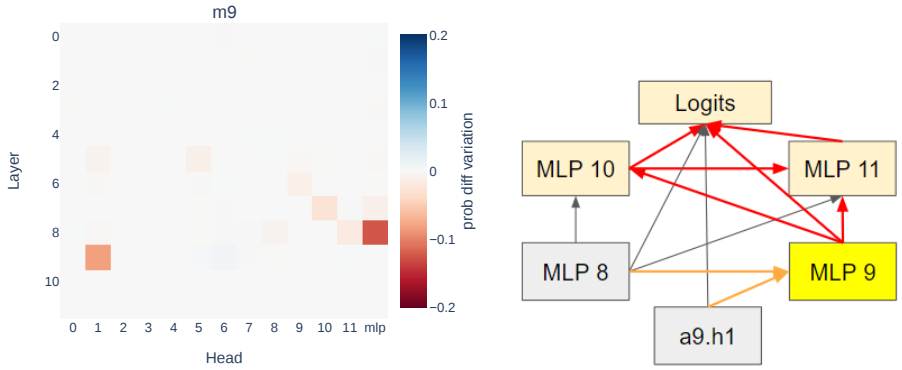

Figure 18: Path Patching Step 4: MLP 9

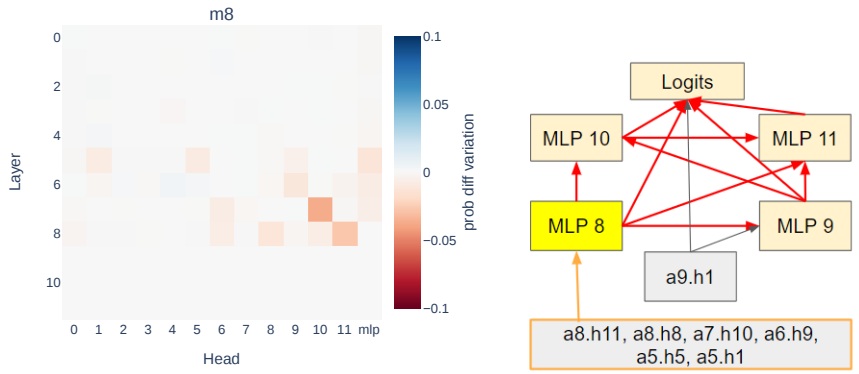

Figure 19: Path Patching Step 5: MLP 8

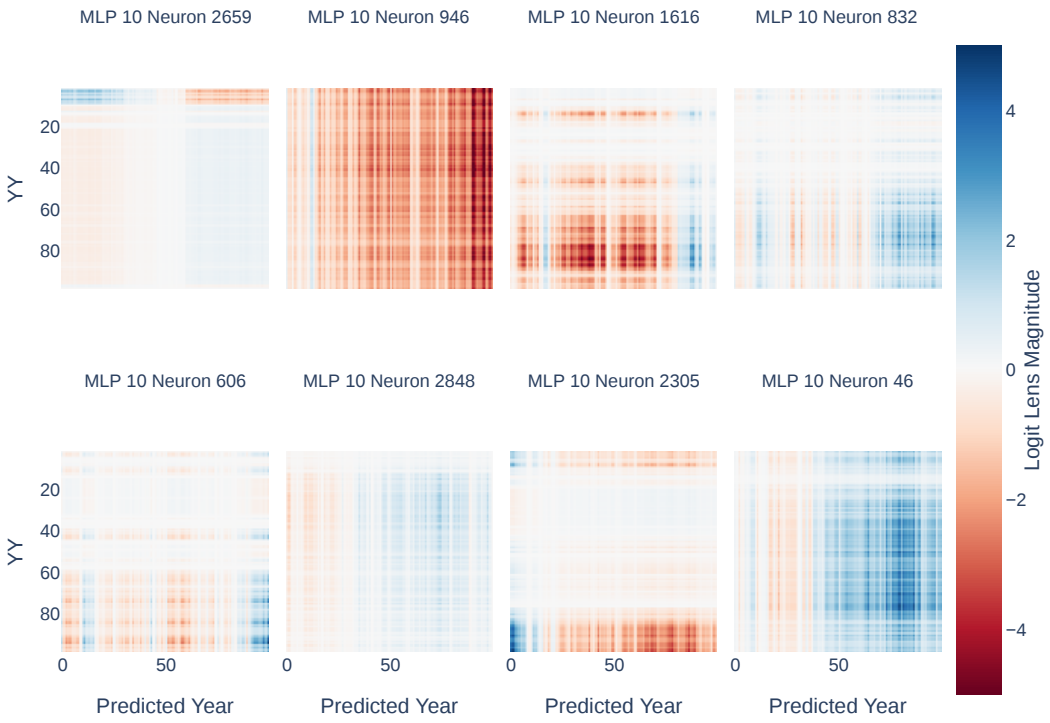

Figure 20: Neuron contributions for each MLP 10 neuron in the top 4-11. Neurons ordered by importance, left-to-right, top-to-bottom. Blue indicates that the neuron upweights a certain predicted year, given a starting year YY, while red indicates downweighting.

At this juncture, it might be appropriate to ask which nodes in the graph most influence the circuit via a9.h1. However, we skip this node because, as we explain in the circuit semantics section (Section 3.3), the attention heads are acting together separately from the MLPs, performing different roles. Moreover, when analyzing attention heads, we must consider what influences their queries, keys, and values separately, a complicated task best avoided in the MLP-centric analysis. In Appendix B, we explore in greater detail the nodes that contribute to such attention heads.

Instead, we complete our circuit-finding section by finding nodes that contribute to the circuit via MLP 8. This reveals (Figure 19) the heads that identify YY, completing our initial circuit investigations.

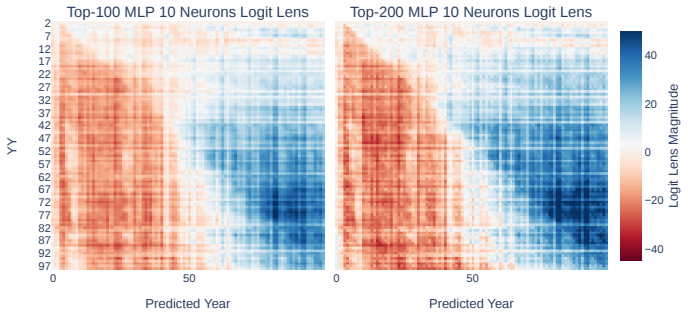

Figure 21: Neuron contributions of the top-100 (left) and 200 (right) neurons in MLP 10. Blue indicates that the neuron upweights a certain predicted year, given a starting year YY, while red indicates downweighting.

D MLP 10 Neuron Contributions

In this section, we display the contributions of the top 10 most important neurons of MLP 10, found in Figure 20. Many neurons' contributions are relatively constant across YY; e.g. the 4th most important neuron always upweights later years. Others differ across YY but not predicted year; the 10th most important neuron downweights all years for the last 10 years or so, where correct answers are very few generally. The 3rd most important neuron varies in both dimensions, having 0 contribution for years YY from around 10 to 50, but a distinct pattern for all other YY. Only the first few neurons are very intense in color, as we have fixed the range of the color scale: these neurons are the most important because they cause the greatest changes when they are patched.

Combining these contributions rapidly produces patterns resembling those of the MLP as a whole. We see this weakly by viewing the top-10 neurons' contributions, but more strongly in the top-100 or 200 (of 3072) neurons (Figure 21). In these logit lens diagrams, there is a consistent increase in logit lens magnitude between YY and YY+1, for a given start year YY.

E Logit Lens vs. Direct Effects via Path Patching

We use the logit lens throughout this paper for consistency, both internally and with prior work. However, the logit lens has flaws: in reality, the residual stream is normalized using layer normalization prior to being transformed into the logits. This nonlinearity could in theory cause the scaling of logit lens to be misleading, perhaps in such a way that affects our conclusions. In this section, we will show that the logit lens results can be recreated with another technique, direct effects via path patching, that avoids this flaw. The two techniques fortunately yield the same insights.

Using the logit lens, we would normally multiply our component of interest's output by the unembedding matrix. To measure direct effects, we take the unpatched, original model's year logits, over each YY, as a baseline. We then patch the direct path between our component and the logits with the 01-dataset, and again record the logits. The difference between the unpatched logits, and the logits when we patch (ablate) our component of interest, reveals the direct effect that said component had. This approach eliminates one major concern of the logit lens: it yields the difference of two sets of logits, which were both produced with layer normalization, and which has interpretable units.

We can perform our neuron-level logit lens experiments by instead patching the direct paths to the top-3 neurons of MLP 10 (Figure 22). The results are essentially identical to the logit lens results, though they differ in magnitude. We can also sum these direct effects, as we summed the logit lens outputs; again, results differ from those of the logit lens only in magnitude (Figure 23). Finally, we can also patch all top-10 neurons as a group, and view their direct effects; these results are identical to those of the summed direct effects (Figure 23). This suggests that our summed logit lens approach genuinely reflected the direct effects that these neurons have on the logits.

We conclude that the results given by the logit lens and those given by direct effects are largely similar, so concerns about the logit lens are not dire. However, we note that all of our logit lens results (including those for entire MLPs and attention heads) are reproducible using direct effects.